

Note on Data Encryption
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Contents.

1	Introduction	1
1.1	Notions of abstract algebra	1

– 1 – Introduction.

Modern computer science relies a lot on security, either via physical solutions or software ones. In this notes we focus on the latter; we mainly cover the most commonly used algorithm in cryptography, as well as some codes used for communications.

In the remainder of this section we provide the reader with some basic definitions of abstract algebra, following with some notion of projective geometry, as these will be at the core of our discussion.

– 1.1 – Notions of abstract algebra.

Here we provide just some basic definitions and theorems, the interested reader is referred to more authorial sources¹.

Let G be a set, and let $+$ denote a binary operation on the elements in G ; then, we say that the pair $(G, +)$ is a *group* if the following hold:

1. There exists an element $e \in G$ such that, $\forall a \in G$ one has that $e + a = a + e = a$,
2. For any element $a \in G$, $\exists -a$ such that $a + (-a) = -a + a = e$, and
3. For any three element $a, b, c \in G$ one has that $(a + b) + c = a + (b + c)$.

Remark: Although we give the definition using $+$ to indicate the group operation, this could differ from the usual addition.

A group for which $+$ is commutative, i.e, $a + b = b + a$, is called *abelian*.

Let $(G, +)$ be a group and let $a \in G$. We denote by a^n the n -th power of a , and denote the set of all powers of a by $\langle a \rangle$. If it holds that $G = \langle a \rangle$, we say that G is a cyclic group. A subset H of G such that $(H, +)$ is a group, is called sub-group of G , and is said proper sub-group if $H \neq G$.

The following holds for cyclic groups:

Proposition: Every cyclic group is abelian. Additionally, every subgroup a cyclic group is cyclic.

Given a group its *order*, denoted as $|G|$, is given by the number of elements in it, therefore we can have finite groups and infinite groups. For finite groups the following holds:

Theorem (Langrange’s theorem for groups): *Let $(G, +)$ be a group and let $(H, +)$ be one of its sub-groups. Then, it holds that $|H|$ divides $|G|$.*

¹A good starting point is Thomas W. Judson’s “Abstract Algebra: Theory and Applications”.

Section 1 – Introduction

We now move on with defining another abstract structure, but before doing so, we would like to point out that groups are intensively studied and a whole field of abstract algebra is built around them.

Following the same approach used for groups, let R be a set and let $+$ and $*$ be two binary operations on the elements of R . We say that $(R, +, *)$ is a *ring* if the following hold:

1. $(R, +)$ is an abelian group,
2. There exists $e' \in R$ such that $e' * a = a * e' = a, \forall a \in G$,
3. For any $a, b, c \in R$ one has $(a * b) * c = a * (b * c)$; that is, $*$ is associative, and
4. For all $a, b, c \in R$ one has $a * (b + c) = a * b + a * c$; that is, $*$ distributes over $+$.

All the properties and concepts here introduced for groups also hold for ring. That is, if $*$ is commutative the ring is called abelian, a subset A of R is called sub-ring if $(A, +, *)$ is a ring, and R is said to be cyclic if there exists $a \in R$ such that $R = \langle a \rangle$.